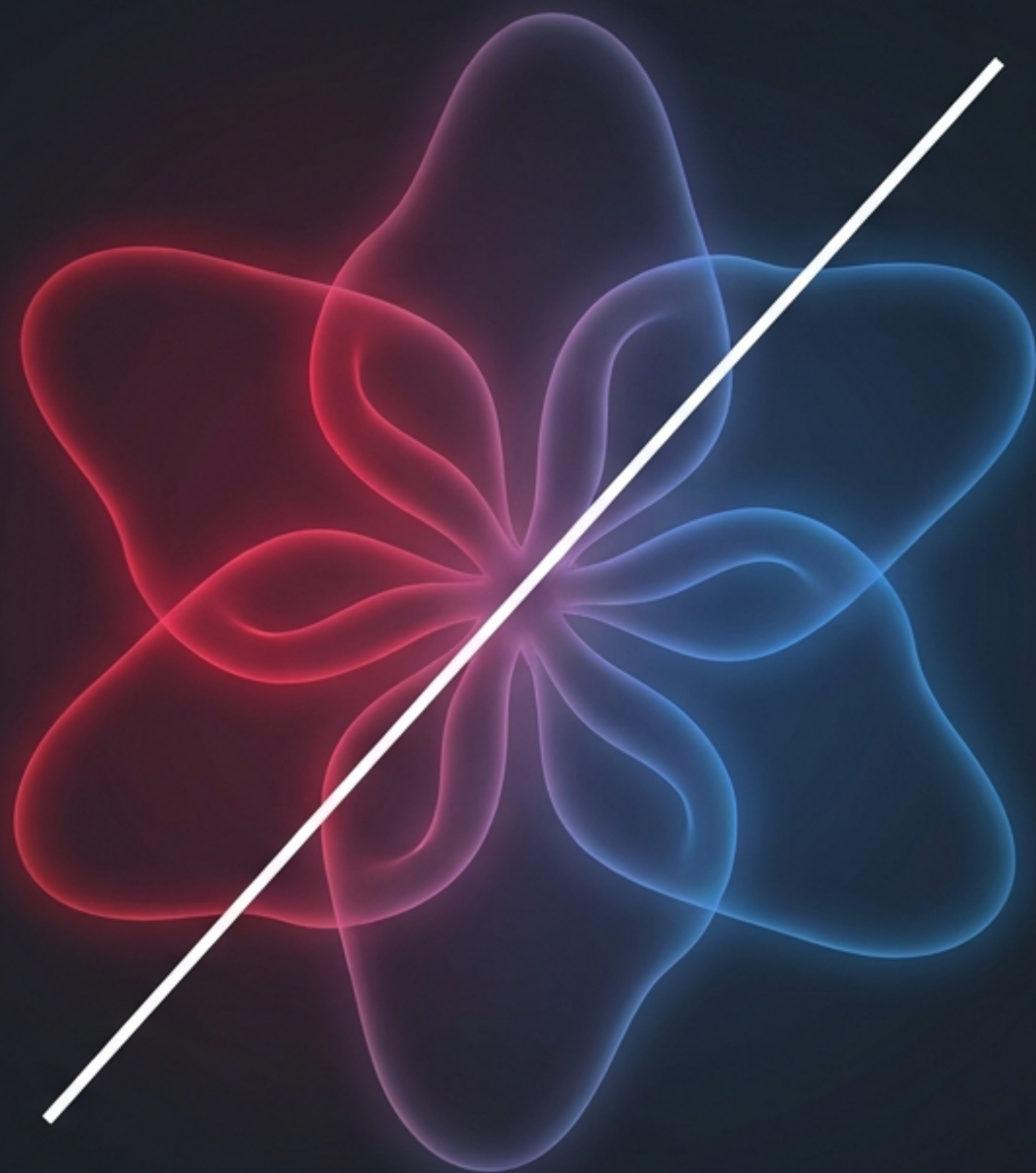


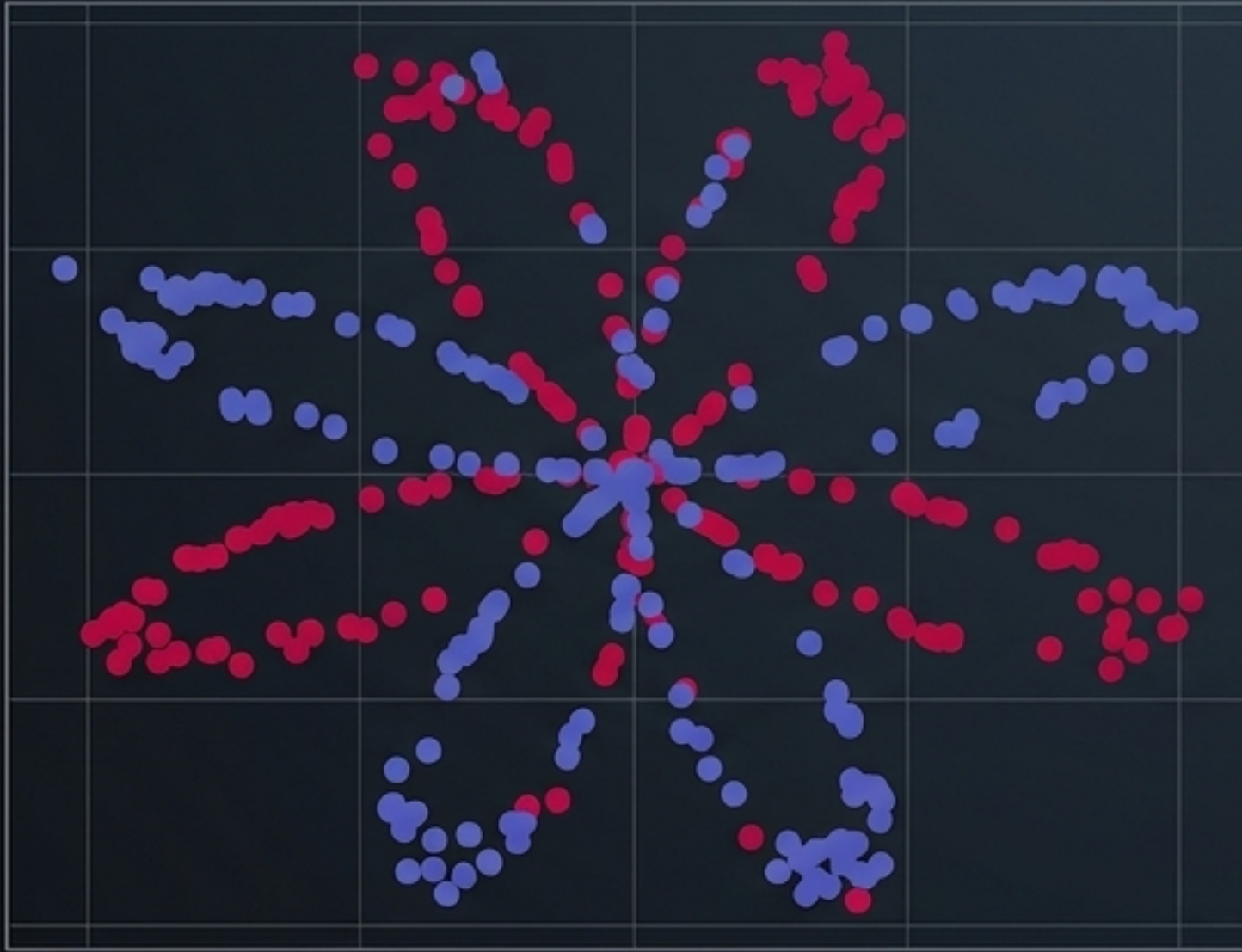
Mastering Non-Linearity

Building a Planar Data
Neural Network from
Scratch

CS4085 Deep Learning | Effat University
Authors: Danya Alshurafa & Sedra Massalha
Instructor: Dr. Naila Marir

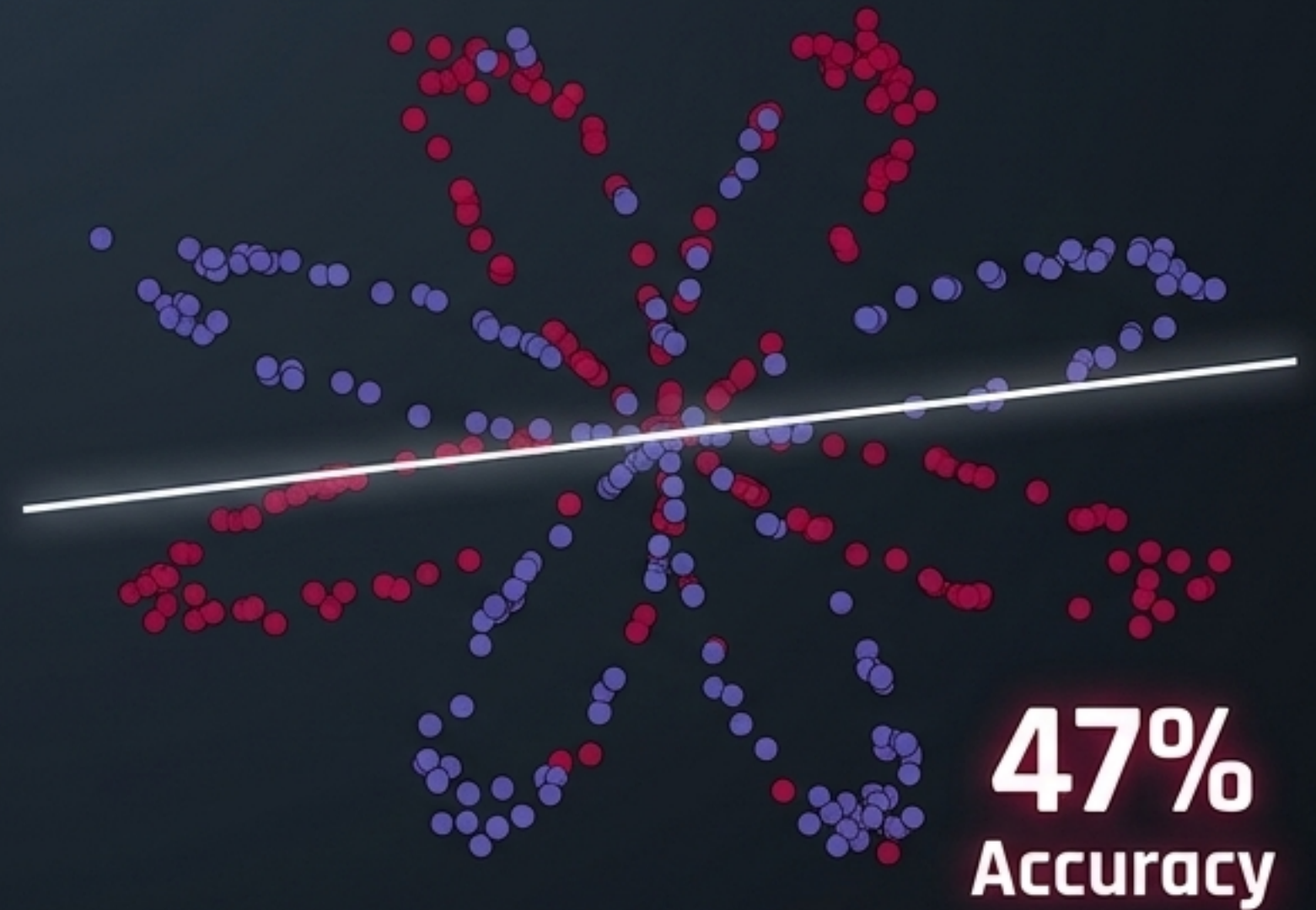


The Challenge



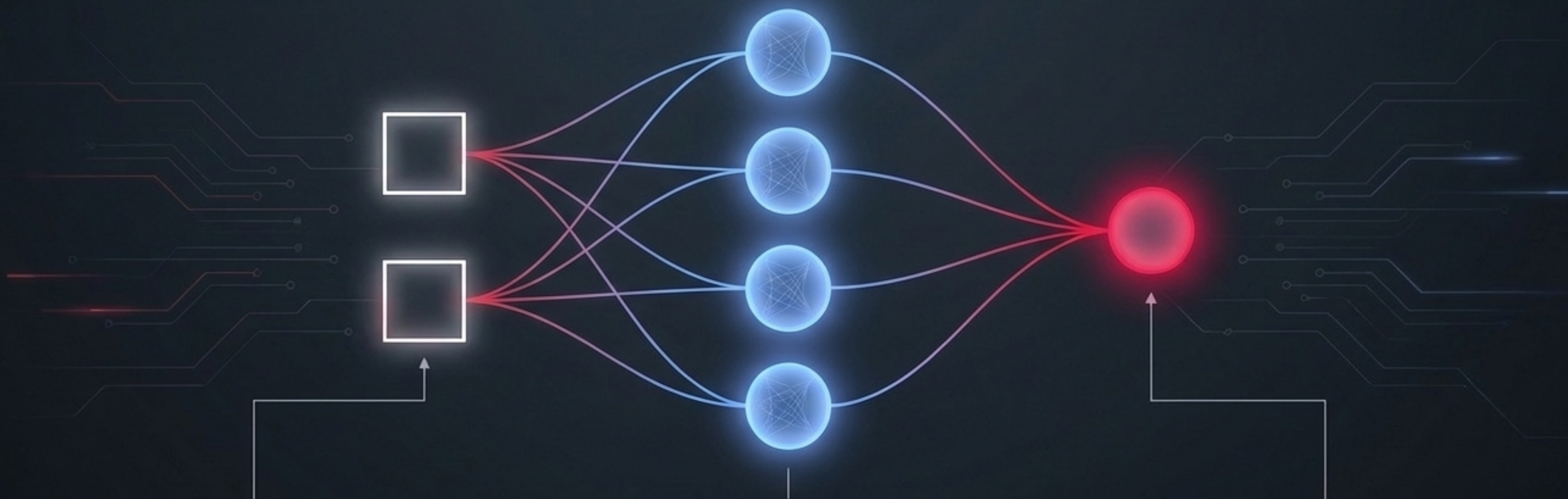
Real-world data is organic and intertwined.

The Linear Failure



Logistic regression assumes a linear world. When faced with the non-linear reality of the flower dataset, it performs worse than a coin toss.

The Architecture of Flexibility



Input Layer ($n_x = 2$)

Raw coordinates (x_1, x_2) captured from the grid.

Hidden Layer ($n_h = 4$)

$$\mathbf{A}^{[1]} = \tanh(\mathbf{Z}^{[1]})$$

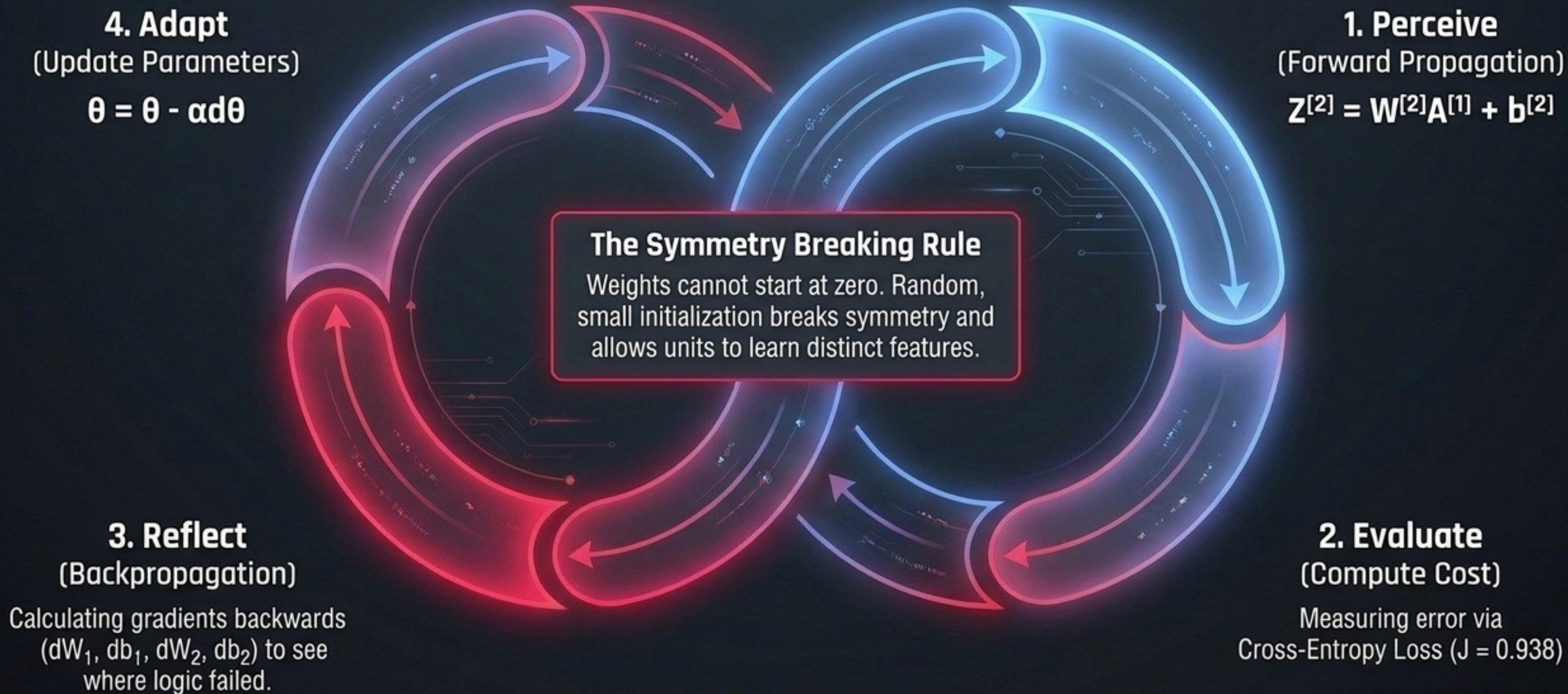
Applying the non-linear tanh activation function allows the network to combine multiple boundaries and bend its perspective to map curves.

Output Layer ($n_y = 1$)

$$\mathbf{A}^{[2]} = \sigma(\mathbf{Z}^{[2]})$$

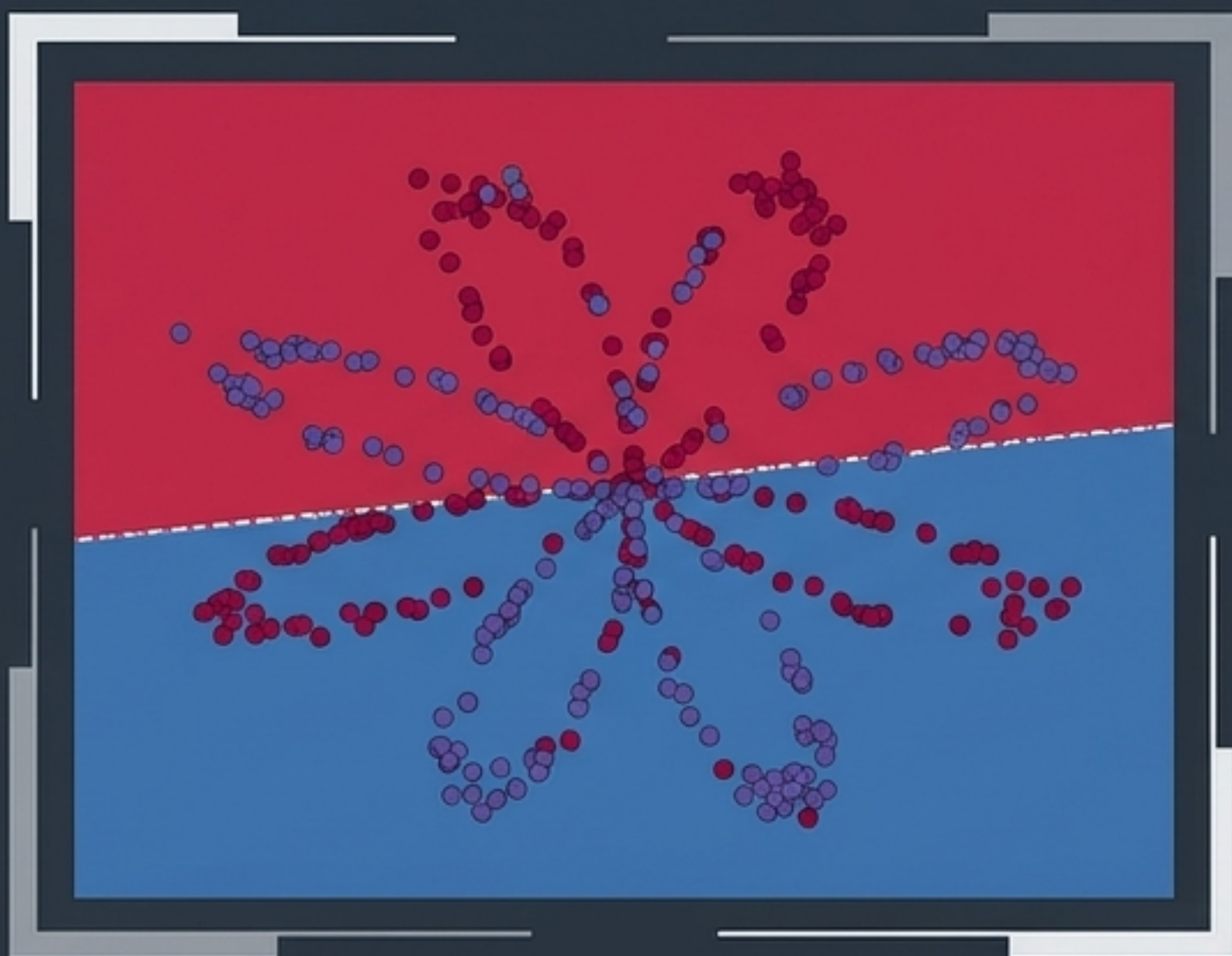
The sigmoid function outputs a final probability, cleanly separating Class 0 (crimson) from Class 1 (spectral blue) at the 0.5 threshold.

The Engine of Learning



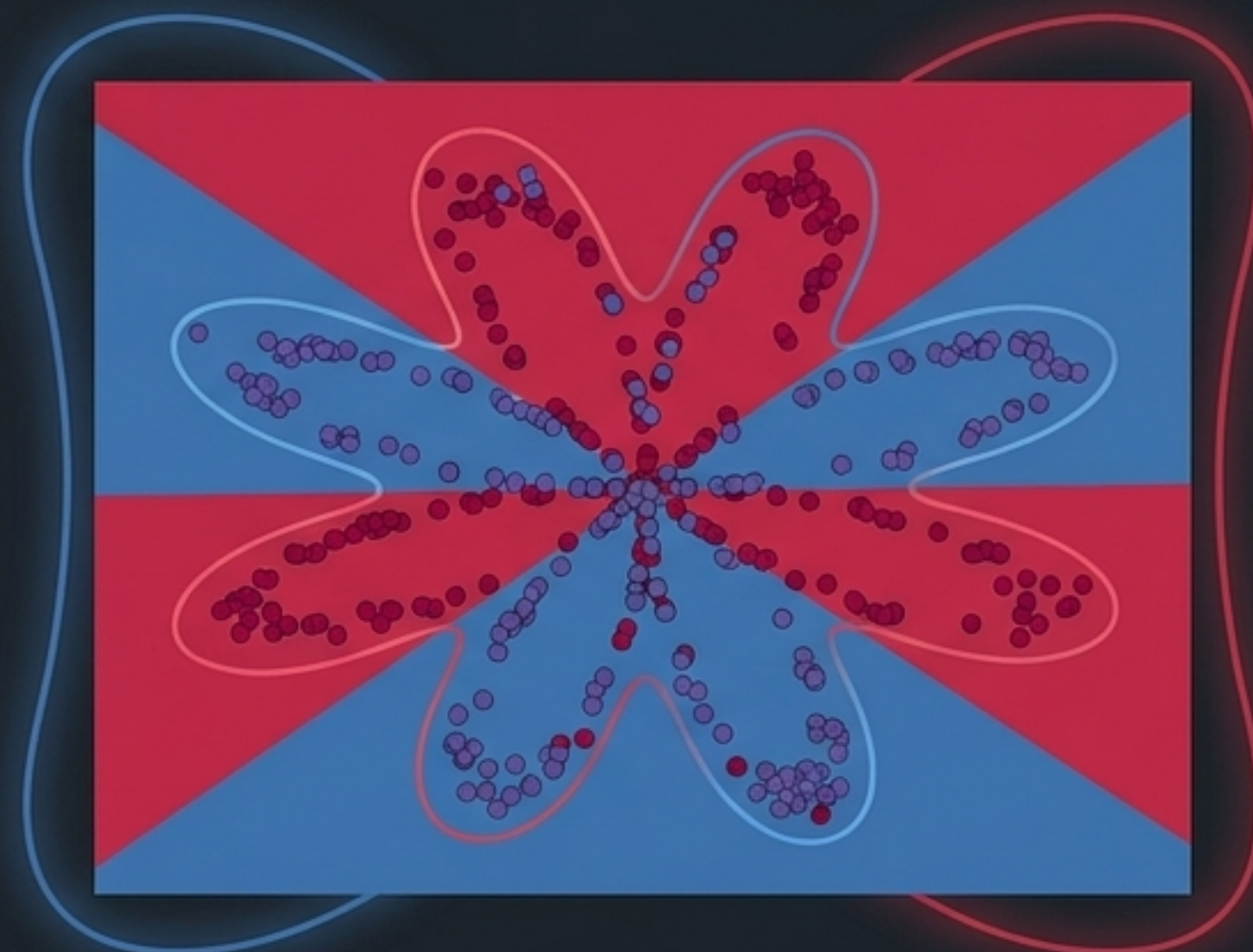
The Rigid Baseline

47% Accuracy



The Organic Solution

90% Accuracy

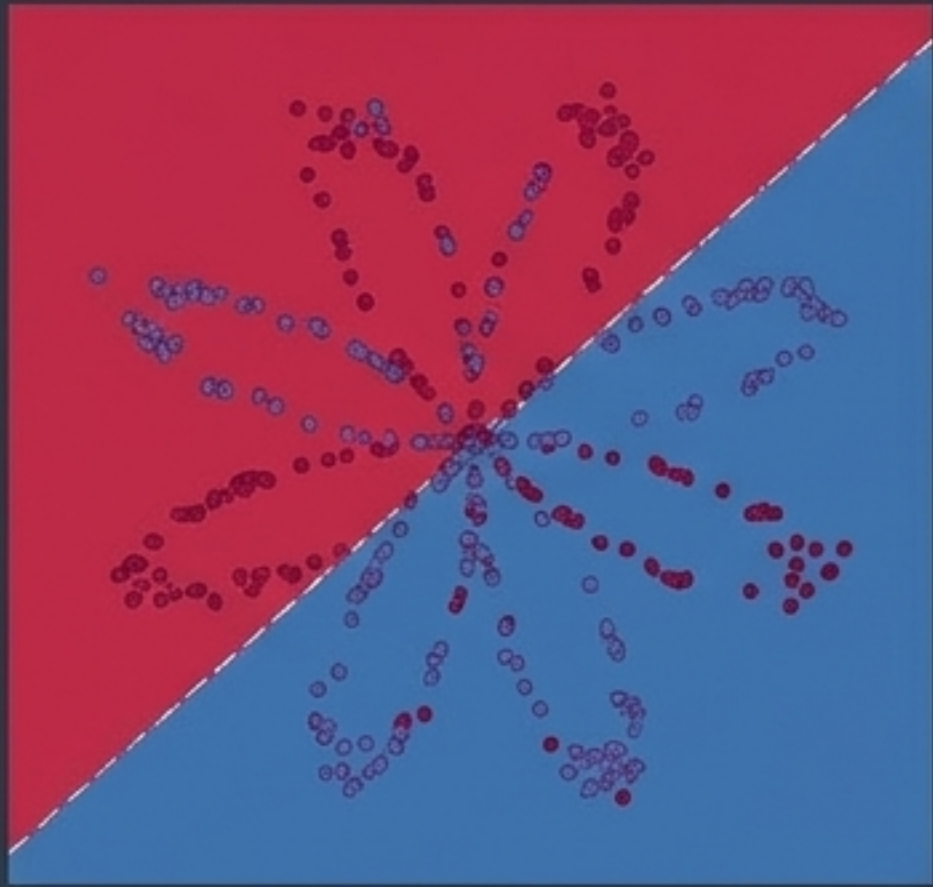


By adding just 4 hidden units and a non-linear activation function, the model successfully learns the **complex, natural shape** of the environment.

The Goldilocks Spectrum

Underfitting

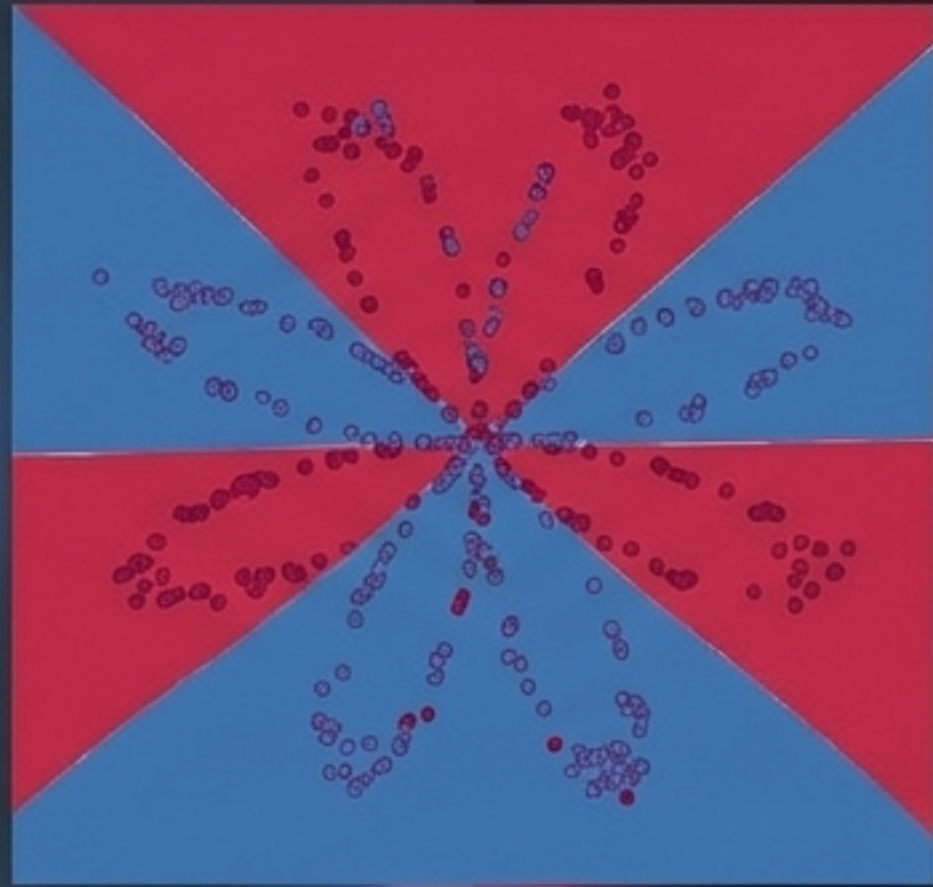
$n_h = 1$ | Accuracy: 67.5%



Too simple. The boundary remains mostly linear, failing to capture the petals.

The Goldilocks Zone

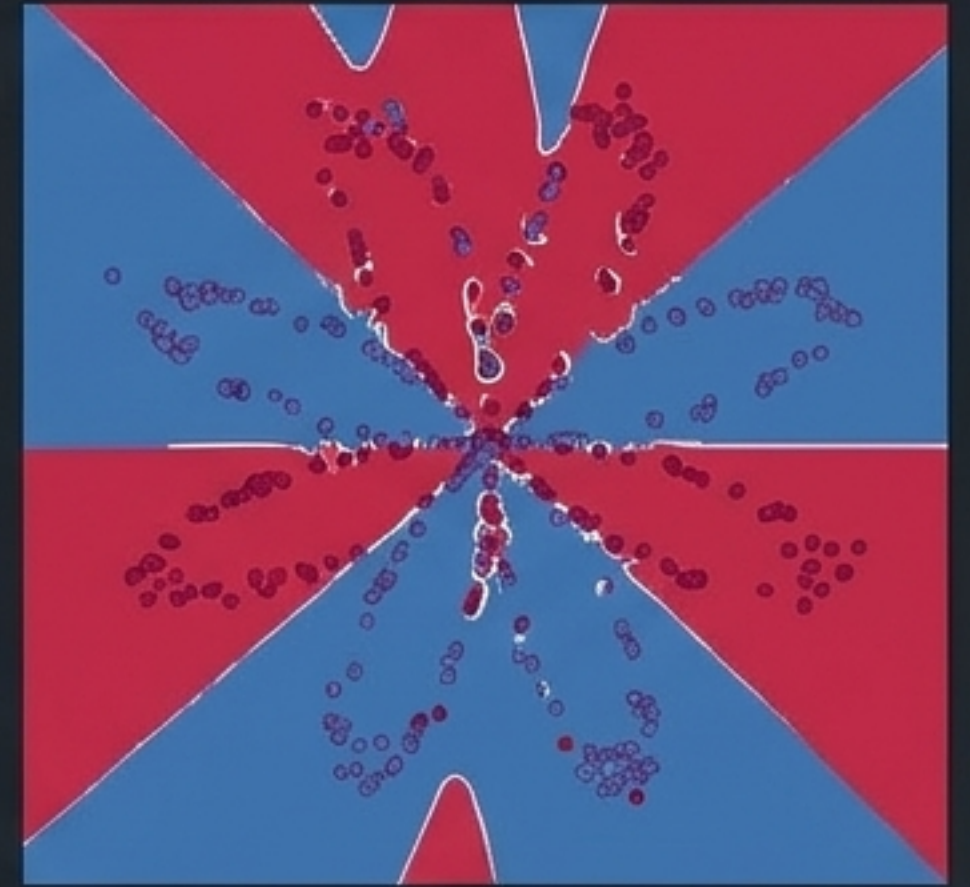
$n_h = 5$ | Accuracy: 91.25%



Optimal complexity. The boundary captures the true underlying pattern smoothly without memorizing outliers.

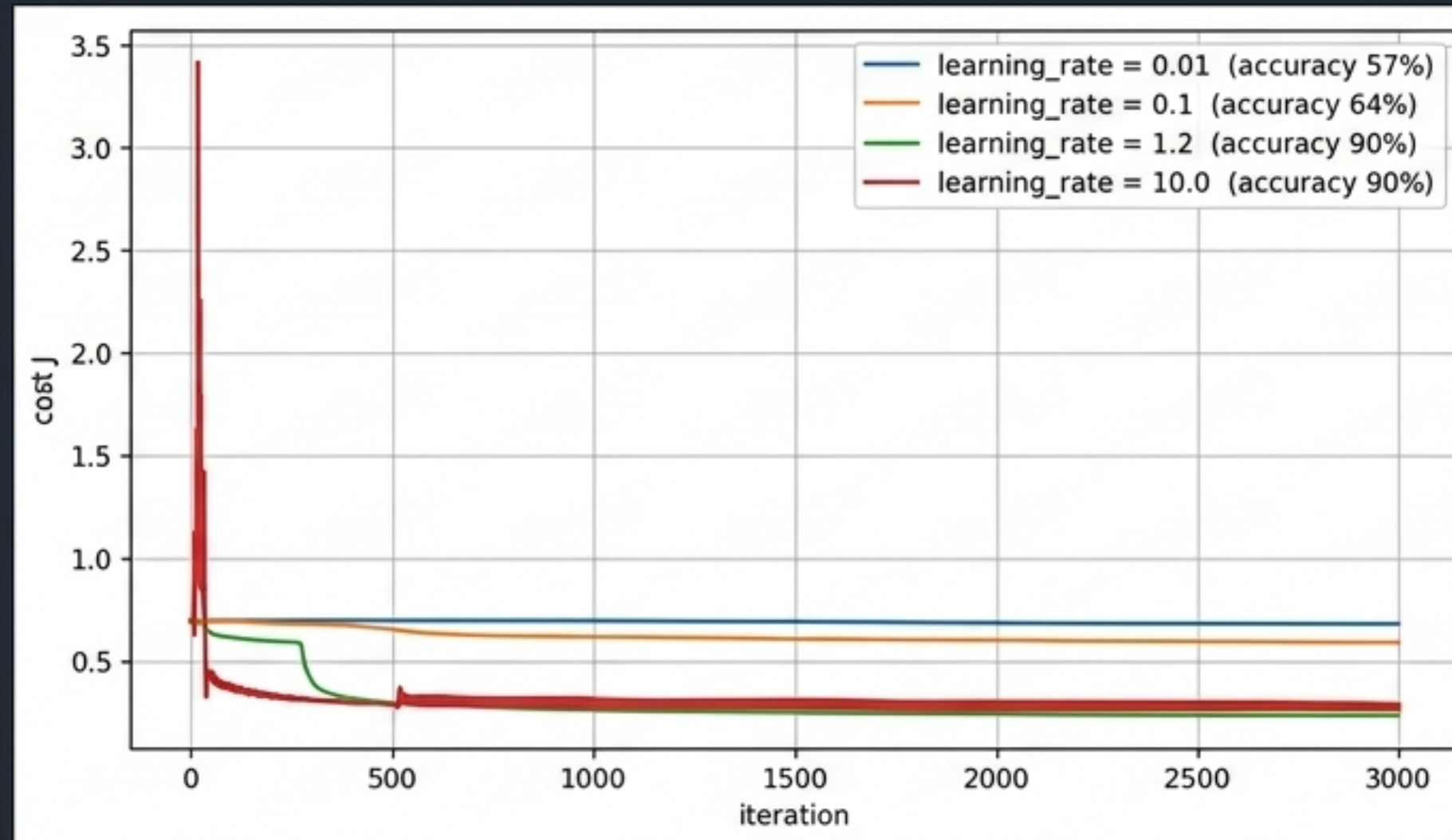
Overfitting

$n_h = 50$ | Accuracy: 90.5%



Too complex. The boundary becomes jagged and erratic, memorizing the noise of individual training points rather than the true biological shape.

The Pace of Learning



— The Sluggish Descent ($\alpha = 0.01$)

Barely learns; flatlines at 57% accuracy. Updates are too small to reach the minimum.

— The Optimal Glide ($\alpha = 1.2$)

Fast, stable convergence reaching 90% accuracy.

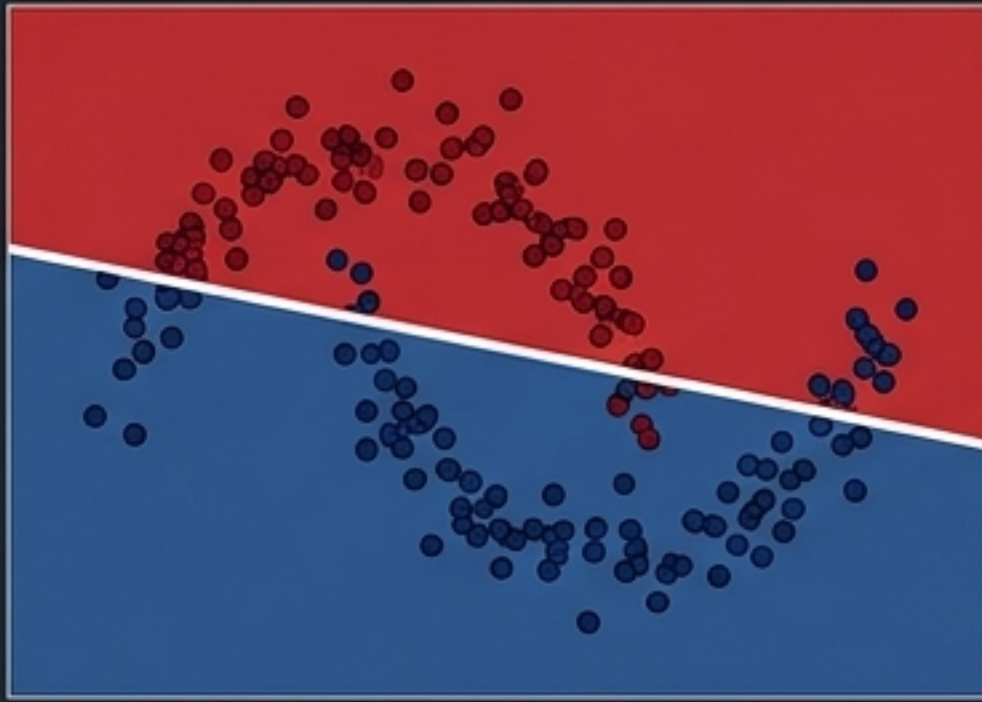
— The Chaotic Leap ($\alpha = 10.0$)

Erratic, overshoots the minimum repeatedly. High volatility.

Generalization: Beyond the Flower

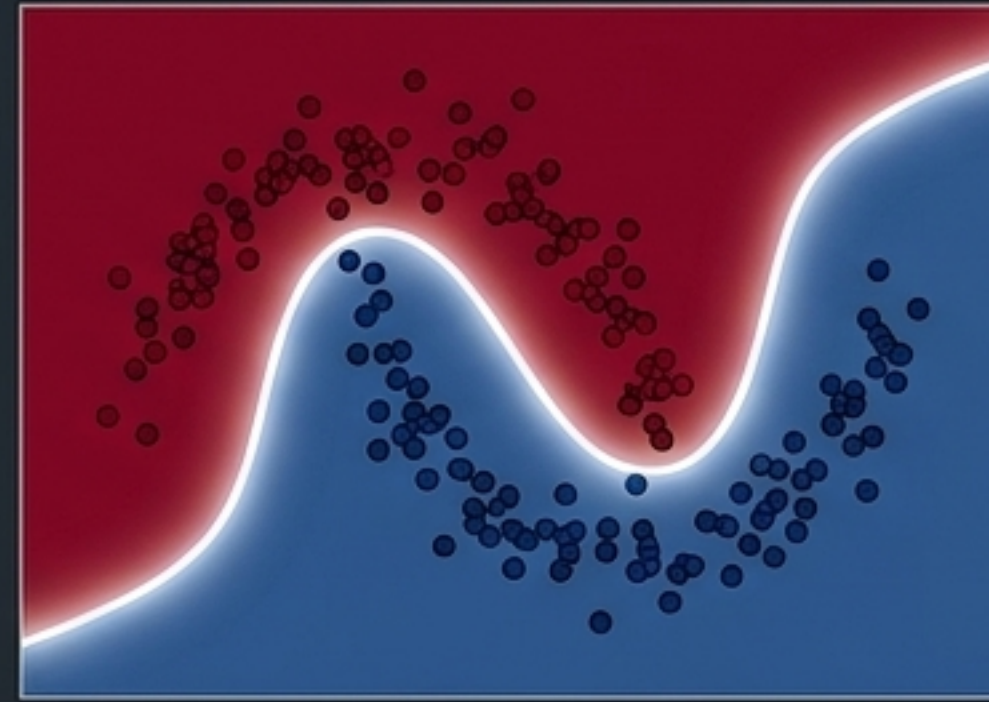
Testing the tuned model ($n_h=5$) on an entirely new topology: interwoven crescents.

Logistic Regression Attempt



Logistic Regression Attempt
85% Accuracy

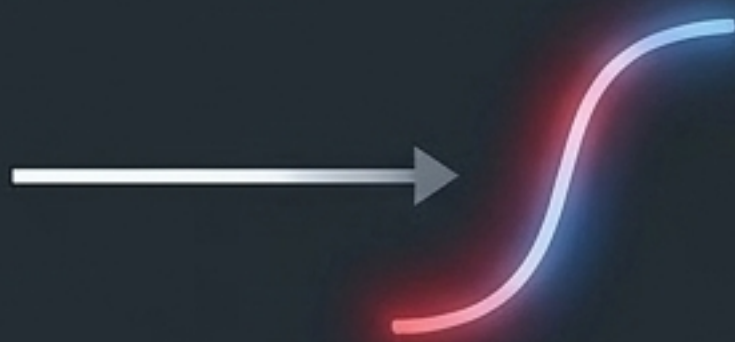
Neural Network Triumph



Neural Network Triumph
99% Accuracy

The greatest advantage of the hidden layer is its generalized adaptability to diverse geometric entanglements.

Synthesis & Key Takeaways



Non-Linearity Shapes Reality

Real-world data doesn't fit in boxes. Activation functions like tanh act as lenses, allowing algorithms to bend their perspective to match complex, organic structures.



Capacity Demands Restraint

Increasing network size yields diminishing returns. Unchecked complexity leads to overfitting—memorizing the noise of the past rather than predicting the signal of the future.



Optimization Requires Asymmetry

Perfect uniformity is the enemy of learning. Breaking symmetry via random initialization is fundamentally required for a network to evolve, adapt, and distribute its learning capacity.